Project Description – Project Proposals in the Area of Scientific Library Services and Information Systems (LIS)

LIS Funding Programme or Call]	"Information Infrastructures for Research Data"
[Project title]	**World Heat Flow Portal**
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Follow-up application (Phase 2) to the *World Heat Flow Database* project (project number 491795283.)

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1. Starting Point

1.1 …in a nutshell

The World Heat Flow Portal (WHFP) project, a continuation of the World Heat Flow Database (WHDB; 04/2022 – 03/2025), is dedicated to consolidating and enhance the current research data infrastructure (RDI) for global terrestrial heat-flow data. This project is designed to provide the geoscientific community with efficient and open access to and utilization of over a century's worth of research and data collection. The project emphasizes the improvement and expansion of the current RDI and web-based data portal, focusing on increased resilience, interoperability, and user experience. Key initiatives include transitioning the data publication process from the current prototype to a fully semi-automated system, expanding the database with new data types and enhanced metadata standards (esp. for marine heat flow data), and strengthening both front-end and back-end functionalities. These enhancements will include additional API representations that build on the current RESTful implementation and conform to alternative community expectations (e.g., STAC, OGC), advanced analytical tools, educational resources, and community-driven data and metadata enrichment. A crucial aspect of the WHFP is the integration of marine heat-flow data, bridging the fields of geoscience and oceanography, and establishing standard practices across these disciplines. The project aims to create a comprehensive, one-stop shop for heat flow research, facilitating data sharing and enabling advanced geoscientific interpretations using modern analysis and visualization tools. The final data portal will uniquely allow the joint analysis of estimated heat flow data, underlying thermal data (such as conductivities and temperatures), geological information, and outputs from subsurface numerical modeling studies. This project is supported by the German National RDI consortium for Earth (NFDI4Earth) and leading experts, including the International Heat Flow Commission (IHFC), the International Lithosphere Program (ILP), and major geophysical societies (IASPEI, IUGG, IUGS). The WHFP is committed to delivering quality-controlled, FAIR-compliant data (→ see reference letters attached). We will update, extend, and homogenize the entire Global Heat Flow Database (GHFDB) during the three-year project, ensuring 100% quality-controlled, curated data. After the project's completion, the RDI will be operated and maintained as a permanent service by GFZ.

1.2 State-of-the-art and preliminary work

Why is heat flow important?

Heat flow, or terrestrial heat-flow density, is a measure of the thermal energy transferring from the Earth's core towards its surface, unaffected by near-surface processes. It varies locally and is quantified as the heat-transfer rate across a surface at a specific location per unit area per time. Accurate knowledge of heat flow is essential for understanding the Earth's temperature field, providing insights into the planet's heterogeneous structure, geologic processes, and evolutionary history. It illuminates phenomena such as active rifts, transient responses to glacial cooling events, tectonic structures, rock-type contrasts, salt structures, fluid-heat transport effects, and the fundamental mantle heat flow entering the crust. Determining heat flow accurately requires methods tailored to specific geological settings, often involving corrections for conductive terrestrial heat flow versus heat affected by processes such as advection, convection, or heat refraction. Providing extensive additional metadata is therefore crucial to assess the quality and accuracy of heat flow values. This includes details about the methods used, applied corrections, geographic and tectonic settings, depth intervals, and the involved rock types.

Heat flow can be measured in both continental and marine environments, making it a key parameter for understanding plate-tectonic and geodynamic processes across various scales. It is systematically measured and reported in scientific drilling projects such as the International Ocean Drilling Program (IODP) and the International Continental Scientific Drilling Program (ICDP). Reliable heat-flow data are critical for thermal modeling in resource assessment, geothermal energy exploitation, hydrocarbon maturation, and the safe geological storage of energy and waste. They are also essential for managing long-term subsurface utilization, particularly in densely populated areas and urban centers, where underground usage is strategic in the ongoing energy transition. Without reliable heat-flow data, many scientific analyses and industrial applications lack a crucial input parameter for accurate modeling and validation.

Status Quo of Heat Flow Data, Compilations and Infrastructure

Globally, heat flow data have been systematically compiled since the early 1960s by the International Heat Flow Commission (IHFC; www.ihfc-iugg.org). Despite significant achievements in the 20th century, the IHFC became less active from 2000 to 2020. Their data compilations have become outdated, lagging behind the advancements in data science, database technology, and digitization. By 2020, the IHFC's data compilation, still in tabular form, was based on technological concepts from the 1970s (cf. *Jessop et al., 1976*). **Key issues** identified in joint discussions with the geoscientific community **before Phase 1** (= WHDB project**)** included:

- **Rudimentary Data**: Many data compilations included only basic information on heat flow and location coordinates, lacking sufficient metadata and quality control. This made it challenging to assess data reliability for scientific research and industrial applications.
- **Multiple Undocumented Versions**: The existence of various undocumented heat flow compilations across the internet created confusion within the geoscientific community.
- **Absence of a Modern Data Infrastructure**: There was no modern infrastructure to support the data, limiting accessibility and usability.

The **primary objective of phase 1 was to develop a prototype** of a modern, user-oriented research data and information platform that adheres to FAIR (Findable, Accessible, Interoperable, Reusable; cf. *Wilkinson et al., 2019*) and OPEN data principles. The platform aimed to ensure the provision of quality-controlled and validated data, following accepted metadata standards. Key outcomes from the 1st phase include:

- Establishment of **new standards** for data formats, metadata schemes, quality schemes, and controlled vocabulary codes through a community-driven process.
- **Collection** of requisite documents and primary **literature** for 90% of the heat flow data.
- Development of a **new relational database** for heat flow data.
- Community-driven **recompilation** and **metadata enrichment** for 60% of the global data, including the assignment of persistent identifiers such as DOI, ORCID, IGSN, and ROR.
- Creation of a **prototype** online portal with open-source publication of the platform code, featuring interactive maps, specific geoscientific data visualizations (1D borehole and 2D profiles), and basic analysis and statistical tools.

- Progress in developing a basic **API** for interoperable data usage and implementing a **data-publication workflow** with GFZ Data Services.

Throughout the 1st phase, community engagement and community building were the keys to developing generally accepted metadata and quality schemes and to successfully tackling the quality control of the Global Heat Flow Database. For a detailed overview of the findings of the 1st phase, we refer to the intermediate project report submitted with this proposal.

Scientific needs, development required, and project goals for Phase 2

Building on the prototype established in Phase 1 and achieving a 60% data assessment rate, Phase 2 is crucial for consolidating the technical database and transitioning the data portal from prototype to full deployment. The primary technical aspects include: (1) **enhancing the infrastructure resilience** to ensure stability and robustness for long-term use, (2) further automating the DOI-minting data-publication process to **streamline the data publication workflow**, and (3) **expanding the database** to include additional metadata and quality control procedures, particularly for supporting thermal data, like rock conductivity and temperature. Regarding data aspects, (4) **finalizing the community-driven recompilation** and metadata enrichment of globally unassessed heat flow data (for terrestrial and marine heat flow) and incorporating additional unpublished data (approximately 400 publications) are crucial. This process will be facilitated by the harmonization and standardization of oceanic heat-flow data in collaboration with the oceanographic community and the development of metadata profiles for conductivity and temperature data, which are indispensable supportive information. To support comprehensive and integrated scientific analysis, the following developments are necessary: (5) standardization of global geological and surface data (e.g., crustal composition, elevation models, maps of sedimentation/erosion rates, mean surface temperature, hydrothermal circulation) to appropriate grid scales to enable automated geographic cross-referencing with heat-flow data and associated corrections.

The *World Heat Flow Portal* Project aims to consolidate and finalize the new heat-flow RDI with a modern web interface developed during the 1st phase (the prototype from Phase 1 is to be launched early 2025 at www.heatflow.world). After completion of Phase 2 (this proposal) the heat flow portal will feature 100% quality-controlled, up-to-date, well-documented, and restructured heat-flow data, and establish itself as a standalone reference and a comprehensive resource for heat-flow-related data and information. It will be prepared for its final release and permanent operation at GFZ. The portal will follow accepted metadata standards, ensuring full interoperability with other geoscientific data services and the integration into the national and international landscape of RDIs (e.g., NFDI4Earth). Reflecting the FAIR and open data principles, the database will provide the geoscientific community with a solid foundation to apply new data science approaches for a better understanding of the geosphere. Additionally, the new semi-automatic DOI minting service for newly submitted heat-flow data will be completed.

1.3 Own prior work and representation of the applicant organizations

On the level of library and information services and web-based data infrastructures, the partners GFZ-LIS, TUD and UB are the national developers of modern RDI and repositories for research data with long-standing experiences (e.g., NFDI4Earth, IGSN, PANGAEA, GFZ Data Services). The proposed projects will perfectly fit into and benefit from these national and international activities. The GFZ is Germany's national research Centre for the solid Earth sciences with the mission to deepen the knowledge of the dynamics of the solid Earth, and to develop solutions for grand challenges facing society. MARUM, the Center for Marine Environmental Sciences at the University of Bremen, is a leading research institution dedicated to studying the dynamics of the marine environment and its interactions with the Earth's system. Its research is crucial for understanding climate change, marine resources, and geohazards, providing essential insights for sustainable ocean management and environmental protection. In addition, it is hosting the PANGAEA, which is a CoreTrustSeal-certified data repository.

GFZ-LIS – Library and Information Services

Since 2000, the GFZ Library and Information Services (GFZ-LIS) have been actively involved in various national and international networks, projects, and initiatives related to research data management, data publication, and open science. For example, the “invention” of the DOI for data STD-DOI (Klump et al., 2006), the Research Data Alliance, World Data System, CODATA, re3data - Registry of Research Data Repositories (e.g., Pampel et al., 2023; Kindling et al., 2017), COPDESS, Enabling FAIR Data in the Earth, Space and environmental sciences (Stall et al., 2018), the Global Geodetic Observing System (GGOS) and OneGeochemistry. With GFZ Data Services within LIS, a strong infrastructural unit was built as a key player in research data management (Ulbricht et al., 2016; Elger et al., 2018, Conze et al., 2017). GFZ Data Services is an international research data repository for the Geosciences domain with the first DOI assigned in 2004. In addition to our focus on data from “long tail communities”, we have developed several DOI minting services for larger projects, international services, and networks. Among others, GFZ Data Services is the data publisher for several services of the International Association for Geodesy (ICGEM, IGETS, ISG), the World Stress Map, INTERMAGNET, GEOFON, the Geophysical Instrument Pool Potsdam GIPP, TERENO, EnMAP, the Potsdam Institute for Climate Impact Research PIK, the Specialised Information Service for Geosciences FID GEO (Achterberg et al., 2018, www.fidgeo.de). It also maintains the GFZ IGSN Service and Sample Catalogue (IGSN = International Generic Sample Number) (Conze et al., 2017). Beyond data publication services, we have long-standing experiences in the development of databases for data publication services, registration of DOIs, IGSN, and other persistent identifier and data portals at GFZ (e.g., SPP Earth Shape, SPP 4DMB, Medusa, WHDP), and are serving as a central information point for research data management at GFZ.

GFZ-ETG – Working group ‘Exploration of Thermal Geosystems’

The GFZ working group "Exploration of Thermal Geosystems" (GFZ-ETG) in the Department of Geosystems focuses on geoenergy exploration and geothermal research. Key areas include geothermal projects, heat-flow and petrophysical databases, thermal field modeling, georesource assessment, lithosphere thermal studies, petrophysical rock analysis, borehole temperature logging, and subsurface paleoclimatic analysis. GFZ-ETG has extensive expertise in thermal studies and heat-flow analysis, demonstrated by numerous research projects and publications over the past decade (Fuchs et al., 2015; Fuchs & Balling, 2016a,b; Förster et al., 2016; Norden et al., 2020; Fuchs et al., 2020; Förster et al., 2021; Fuchs et al., 2021; Frick et al., 2022; Fuchs et al., 2022; Norden et al., 2023; Fuchs et al., 2023; Neumann et al., in prep.). GFZ-ETG has conducted thermal analysis on various scales through projects such as SFB 267, CLEAN, DESERT, GeoEnergy, INTERREG4A GeoPower, DSF Geotermi, NEOGENE, GeoFern, GeoTiefWien, GeoPuR, WHDB, and more. The group studies the interaction between geology and thermal fields, focusing on parameters like heat flow and thermal rock properties, using laboratory and borehole data integrated into thermal models. Our work covers global to local scales, including geothermal energy, energy storage, and carbon dioxide and nuclear waste storage. In heat flow research, we have led **global collaborations for reassessing the Global Heat Flow Database** (IHFC webpage). Some members (Ben Norden, Florian Neumann, Elif Pazvantoğlu) serve on the International Heat Flow Commission (IHFC, term 2023–2027). Project PI Sven Fuchs is Vice Chair (since 2023) and custodian (since 2019) of the IHFC and chairs Task Force VIII (2021-2025) on Lithospheric Heat Flow of the International Lithosphere Program (ILP). We have driven the standardization of data and metadata (Fuchs et al., 2021, 2023) and produced many quality-controlled data releases (e.g., Fuchs et al., 2021, 2022; Global Heat Flow Data Assessment Group et al., 2023, 2024; Balkan-Pazvantoğlu et al., 2024; Fuentes-Bustillos et al., 2023; Sidigam et al., 2023). This work gives us deep insights into the data landscape and scientific needs, and our efforts have been well-received, with many researchers supporting the collaborative data revision.

TUD – Chair of Geoinformatics

The research activities of the chair of Geoinformatics at TU Dresden (TUD-GI) focus on geoinformation systems, web-based RDIs, smart environmental monitoring, modeling, and distributed spatial analysis. TUD-GI has extensive expertise in research data management and infrastructures from various projects (Bernard et al., 2014). TUD-GI coordinated the development of a scientific GDI in the GLUES project, which facilitated interdisciplinary data exchange and published scientific results (Eppink, 2012). They developed MetaViz, a tool for exploring metadata and visualizing simulated data provenance (Henzen et al., 2013, 2014). The COLABIS project (BMBF) aimed to create a web-based platform for urban monitoring and early warning systems, integrating diverse data sources (Henzen, 2016). In the EXTRUSO (EU-ESF) TUD-GI developed an information infrastructure for hydrometeorological event analysis (Wiemann et al., 2018). TUD-GI also coordinated GeoKur (BMBF), focusing on data curation and quality assurance during research. This included automated metadata acquisition and linking to ontologies. Most TUD-GI projects involve developing geoportals or web applications, with a strong emphasis on usability, testing, and integrating feedback (Henzen, 2018a,b). In the context of the German NFDI, TUD-GI is actively involved in the NFDI4Earth and Base4NFDI projects. Both projects are coordinated by TUD-GI. TUD-GI builds upon well-established links to national and international networks and initiatives (e.g., AGILE, iEMSs, INSPIRE, GEOSS, EuroSDR, OGC) addressing not only research communities but also standardization and legislation bodies, various administrative bodies, further stakeholders, and industry. TUD-GI specifically contributed to the latest revision of the OGC Web Processing Service specification (WPS2.0); the drafting of the INSPIRE Implementing Rules and supported the design and development of several GDI on national and international level (GDI-DE, SDI Croatia, GDI NRW, GDI Saxony). Lars Bernard (chair of TUD-GI) is a member of the RFII (‘*Rat für Informationsinfrastrukturen*’), established by the German Joint Science Conference (‘*Gemeinsame Wissenschaftskonferenz*’). He is the spokesperson of NFDI4Earth & Base4NFDI.

UB – MARUM

MARUM, the Center for Marine Environmental Sciences at the University of Bremen, is a transdisciplinary research center focused on understanding interactions between the Earth's interior, oceans, and atmosphere. The Marine Geotechnics group specializes in the physical properties of marine sediments and conducts research critical to environmental sustainability, resource management, and geohazard assessment. The group operates two state-of-the-art engineering geological-geotechnical laboratories where they analyze sediment-physical (geotechnical) parameters essential for soil and rock stability, including thermal conductivity. Their research applications include assessing slope stability by static and dynamic loading of water-bearing sediments and rocks, and shear experiments at elevated pressures and temperatures found at active continental margins. The Marine Geotechnics group engages in diverse research projects across various regions, spanning from mid-ocean ridges (e.g., Reykjanes Ridge) to subduction systems (Japan, Mediterranean Ridge, Gibraltar Atc, Marianas), but also marginal seas (North Sea, Baltic Sea, Ligurian Basin) and lakes (Switzerland, Chile). The key focus is on the *in-situ* measurement and long-term monitoring of physical parameters such as pressure, temperature, electrical conductivity, as well as seismicity and tilt. Over the past decades, the Bremen marine heat flow team has also developed an outstanding reputation globally, both concerning their in-situ HF probes and T measurements in so-called CORK borehole observatories as part of IODP. A key tool is the Bremen Heat Flow Probe, a 6-meter instrument used for in-situ temperature and thermal conductivity measurements, featuring 22 thermistors and a heater wire. It can operate autonomously or in real-time to collect thermal data. Miniaturized Temperature Data Loggers (MTL), developed with ANTARES, are used for long-term monitoring of fluid flow and mud volcanism. Additionally, the Shallow Water Free-Fall Cone Penetration Testing (SW-FF CPT) probe is used in waters up to 500 meters deep to assess sediment strength and geotechnical properties, aiding marine structure stability. (Pfender, M. and Villinger, H., 2002, Stegmann et al., 2006). The Marine Geotechnics group's work significantly advances the understanding and management of marine geotechnical processes. Their research supports the sustainable development of marine resources and enhances the ability to predict and mitigate geohazards, providing valuable insights into the dynamic marine environment.

2. Objectives and work programme

2.1 Anticipated total duration of the project

The proposed project, which is the 2nd phase of the development of the World Heat Flow Portal, is planned for a duration of 3 years. We are applying for DFG funding for **36 months**. At the end of the funding period, the GFZ will have implemented the proposed RDI as a permanent GFZ operation and service.

2.2 Objectives

The main objectives of the proposed project Phase 2 are:

Consolidation of results of Phase 1

- Ensuring service and server structure stability through comprehensive testing, container-design, monitoring, failover**,** and database replication system, and a comprehensive metadata management, extended documentation**,** and automatized backup functionality
- Finalization of a semi-automatic module for DOI publication (via GFZ Data Services)
- revision and publication of the controlled vocabulary defined in heat flow metadata and quality scheme and integration of new metadata properties for marine heat flow.
- Finalization of the heat flow data assessment

Strengthening

- Developing concepts and adequate metadata schemes in a community-driven workflow for including first-order heat flow parameters (thermal conductivity and subsurface temperature) and additional spatial 2D grid data on temperature and surface heat flow
- Extending the database to also include: first-order parameters and computational spatial 2D grids of temperature and (surface) heat flow
- Providing user-driven code collection of specific data processing codes (for heat flow, thermal rock properties, temperature) with upload or interlinkage to existing GitHub repositories
- Increasing the resilience of RDI by comprehensive testing and code documentation

2.3 Work program and proposed methods

Work packages and research steps

WP1	Start–End: **M1–M36**

Data portal back end, architecture & interoperability
Lead: GFZ-LIS (Kirsten Elger), involved: TUD
Objectives: (1) to consolidate the technical backbone of our web-based data portal, with (2) an enhanced DOI-minting service. (3) interoperability to other geoscientific data services, and (4) full database documentation.
Description of Work: WP1 comprises several strategic steps aimed at **consolidating** the database and the technical backbone of our web-based data platform **and at enhancing it with new impactful functionalities**. Key initiatives include: (1) improved resilience of the technical infrastructure, (2) improved (semi-automatized) data-publication process, (3) extended database structure, metadata schemes**,** and quality procedures for supportive data. The main aim is the **consolidation of the main database and backbone infrastructure** developed in Phase 1 and an increase of the resilience of RDI by comprehensive testing, a continued integration, strengthening of service and server structure through container-design, monitoring to track system performance, failover and database replication system, and a comprehensive metadata management, extended documentation and automatized backup functionality with recovery plan. One important functionality of the portal is the DOI-generating data publication process, for which we have developed and demonstrated the proof of concept in the 1st phase. We now intend to fully transform this manual tool towards a semi-automatic publication module in which the submitted information is converted into the required metadata formats for an automatized setup of the publication process in GFZ Data Services. The controlled vocabulary defined in our heat-flow metadata and quality scheme will be revised and extended and published according to the SKOS guidelines. Following the user-feedback collected during the stakeholder process in Phase 1, the already implemented functionalities of the webpage, community areas, data and project areas, data submission system, and interconnected literature and author database will be updated. **Scientific strengthening of the heat flow database** infrastructure will be achieved by three main steps; (i) by extending the existing database infrastructure for the basic underlying parameters for heat flow, which are thermal rock properties and subsurface temperature. In addition, the database will be extended (ii) for storing information on computational spatial 2D grids of temperature and (surface) heat flow, produced using numerical (forward and inverse) or statistical (ML/AI) computer models. This requires a modification of the upload and download functionality. The required concepts and adequate metadata schemes are again developed in a community-driven workflow in WP3 will enable interoperability with other platforms. As for heat flow, the extended data are connected to the respective literature reference (DOI where available), the respective authors and their institutes (ORCID and ROR where available), to respective rock samples (IGSN where available). The graphical user interface (WP2) will be extended to display these new selector variables. Our project community functionality will be extended to set up a (iii) user-driven code collection of specific data processing codes (for heat flow, thermal rock properties, temperature) with upload or interlinkage functionality to existing GitHub. The page implements a status system of code verification, where verified codes can be used for global database corrections.
Tasks (responsible project partners) T1.1 – Consolidation of the main database and backbone infrastructure T1.2 – Extended testing, code improvement and comprehensive documentation (GFZ) T1.3 – Transformation from manual to semi-automatic data publication tool (GFZ) T1.4 – Publication of controlled vocabulary via a vocabulary registration service (GFZ) T1.5 –Update and extension of the present database, upload and download functionality for thermal properties

and temperature (GFZ) T1.6 – Update and extension of the present database, upload and download functionality for 2D spatial computation grid data (GFZ) T1.7 – Cross-check for data correction needs; implementation of automatized algorithm (GFZ)
WP Deliverables D1.1 – Semi-automatic data publication tool (M18) D1.2 – Publication of controlled vocabulary (M24) D1.3 – Documentation (M33)

WP2	Start–End: **M1–M36**
Data portal front end, exploration and visualization & mapping	
Lead: TUD (Stephan Mäs), involved: GFZ	
Objectives: consolidation and strengthening the web-based front-end for the data portal, including the user interface, functionalities, analysis, visualizations, and mapping functionalities.	
Description of Work: WP2 comprises several strategic steps to **consolidate and improve** the web-based data platform. Key initiatives include: (1) API conformance, (2) improved analytical tools, (3) expanded educational resources, (4) plugin development, and (5) a UI update. **API conformance** will be ensured, in particular to STAC standard, Open API, and the latest relevant OGC API features for the heat flow (point) data itself (OGC standard not finalized in Phase 1). This will facilitate interoperability with various geospatial data platforms. Comprehensive API documentation and developer guides will be developed to streamline integration with external systems. Continuous monitoring of industry trends and standards updates will ensure ongoing improvement and interoperability. The implemented **visual analytical tools** like the 2D data profile plots for boreholes (computed heat-flow and temperature-depth profiles) and cross sections (computed heat-flow profiles) will be **technically improved and methodologically enhanced**. The enhancement includes (i) validation mechanisms to ensure the accuracy of these analytical tools, (ii) the user-driven implementation of spatial or parameter data required for the computation, and (iii) updated analytical algorithms to reflect advancements in heat flow processing techniques, and (iv) improved usability. Feedback sessions will be established to refine and optimize these tools based on user feedback and needs. **Educational resources** will be generated (WP5) and implemented into the data portal. This will include tutorials and webinars to provide use-case examples and real-world datasets for hands-on learning of the platform functionalities and the possible heat-flow data analysis. To increase user engagement and to strengthen the practice around data analysis and visualization through collaborative projects and shared repositories, we will develop an education tutorial **for Jupyter Notebook integration** for Python-based heat-flow processing. Amongst others, interoperability and reuse of our data will be supported with the development of a **GIS plugin** (ArcMap or QGIS) to implement our data into professional GIS applications. Additionally, user-friendly documentation and tutorials for the implementation and access of our data via plugin or API will be provided. Finally, we will evaluate our user-centered design principles to **refresh and update the UI**, focusing on intuitive navigation, visual appeal, and enhanced user experience. Interactive elements such as context menus and time sliders facilitate data exploration and analysis will be implemented. The refresh will be aligned with usability testing to gather feedback for iterative improvements.	
Tasks (responsible project partners) T2.1 – Ensuring API conformance (TUD) T2.2 – Improving analytical and statistical tools, consolidating visualization and interpolation functions (TUD) T2.3 – Developing education tutorials for Jupyter Notebook integration (TUD)	

T2.4 – Programming QGIS plugin (TUD) T2.5 – Refreshing and updating UI (TUD) T2.6 – Completing documentation (TUD+GFZ)
WP Deliverables D2.1 – GIS plugin (M6) | D2.2 – Consolidated data portal and user interface (M18) | D2.3 – Education tutorial (M12) | D2.4 – Documentation (M18)

WP3	Start–End: M1–M36
Continental thermal data: management and collaborative enrichment	
Lead: GFZ-ETG (Sven Fuchs); involved: TU Dresden, collaborative network	
Objectives: Complete the assessment of the present continental databases towards the new metadata standards using the international collaborative network.	

Description of Work:

The primary objective of WP 3 is to **complete the assessment of the continental/onshore section of the IHFC Global Heat Flow Database** based on the metadata standards and quality schemes developed in the first phase of this project. Non-assessed historical data will be revised and enriched with missing necessary information sourced from primary literature. This extensive and time-consuming task requires the collection and screening of both primary and secondary literature for each data entry to identify and integrate all relevant information. Flags indicating perturbing effects, quality, and data status, along with additional selector variables, will be added to facilitate faceted search and selection in the graphical user interface (WP4). Our collection of primary literature sources (PDF scans) currently stands at 85%. For a fully authenticated database, we aim to increase this to 100%. We will continue to collect and store reference metadata in separate, related data tables. Numerous supporting scientists from the scientific community (cf. Sections 5.3–5.4) will assist with this substantial workload, leveraging our extensive experience from Phase 1 (detailed in Sections 1.2–1.3). This work is part of our established and ongoing international collaboration. The organizational details of this collaborative approach are outlined in WP5. Additionally, our evaluation will begin with the status of the GHFDB release 2025, expected in Q1/2025 as the final deliverable of WP1 of the 1st phase. We anticipate a quality control assessment rate of approximately 60% of the 100,000 data points. We have already identified over 250 additional publications with relevant, yet unconsidered data that need to be integrated. We will continue to publish well-documented, frequent releases of the database for download and sharing in internationally trusted data repositories.

At the start of the project, we will launch an international and open workshop series focused on the **standardization** of **metadata** and data quality for **thermal rock properties** and borehole **temperature** measurements. These workshops will provide crucial supporting information for our heat flow database. The resulting property and temperature data will form the foundation of an extended database model, also incorporating persistent identifiers. In addition to heat flow and supporting measurements, we will standardize data formats and develop metadata profiles for spatial **heat flow and temperature data grids** from numerical models (forward, inverse, ML/AI models). This will enable the online joint interpretation of measured and modeled thermal data on various scales, integrating data-driven and numerically-driven research methods.

To support this effort, we will **harmonize global geological and surface data sets** (e.g., crustal composition, elevation models, maps of sedimentation/erosion rates, mean surface temperature data) to appropriate grid scales and make them available. This will facilitate automatic geographically-based cross-checks with stored heat-flow data. Utilizing these datasets and our metadata flags, we will implement **analysis and correction functions** for perturbing effects (e.g., climate, topography). We also plan to develop **data products to provide global recommendations** on data revisions, such as identifying data that should be corrected for paleoclimate effects based on global glaciation maps and climate models.

As a collaboration with the WHDP and FID GEO, we plan to digitally republish old original publications that are only available in analog form. The digital republication includes the clarification of copyrights (by FID GEO staff), digitalization, and online publication (with DOI) via Geo-LEO*e-docs*, the dedicated FID GEO text repository. Beyond collaborative and community-driven standardization, we will engage with the heat-flow community through various initiatives. We will organize **international workshops** focused on developing standards, schemes, and procedures, as well as **summer schools** aimed at training and knowledge transfer for geoscientists who generate or use heat-flow data. These events will take place during the early to mid-project phases. Additionally, we will employ online **questionnaires** to continually assess the needs and demands of the community and **stakeholders**. This feedback will help us enhance the usability and capabilities of the information tools and services included in our RDI.

Tasks (responsible project partners)
T3.1 – Monitoring and evaluation of the geoscience community expectations and requirements, and development and maintenance of the collaborative community approach (GFZ + network)
T3.2 – Community-driven recompilation and metadata enrichment of the final 40% of the continental heat-flow data and further not-yet-implemented data (ca. additional 250 publications), including the extension of reference, author, and institution sub-database (GFZ + network)
T3.3 – Development of metadata profiles, quality classes, and data classification for rock thermal properties, temperature measurement data, and spatial computational heat-flow grids (GFZ + collaborative network)
T3.4 – Harmonization of global geological data grids for cross-checks and portal utilization
T3.5 – Cross-check for data-correction needs and implementation of an automatized correction algorithm for environmental perturbation effects (GFZ)
T3.6 – Releasing/Republication of historical continental heat-flow publications as digital copy; clarifying the copyright status; making pdf available (FID GEO collaboration) (UB)

WP Deliverables
D3.1 – New metadata and quality classification for property and temperature data (M12)
D3.2 – Extended database structure for property and temperature data (M15)
D3.3 – Metadata schema and DB extension for computational heat flow and temperature grids (M24)
D3.4 – Release literature files with an open license (M24)
D3.5 – Automatized geodata-driven database correction procedure (M30)

WP4	Start–End: M1–M36
Marine thermal data: management and collaborative enrichment	
Lead: UB (Achim Kopf, n.n.); involved: GFZ, collaborative network	

Objectives: Connecting geoscience and ocean sciences (oceanography) and completing the marine heat-flow data assessment according to the new metadata standards developed by the international collaborative network.

Description:

We will utilize project workshops to foster community building between **geosciences and oceanography**, specifically focusing on heat flow data at the interface between the solid Earth and the oceans. This includes supporting preparations and coordinating **standardization efforts** to establish heat-flow observations as standard parameters for oceanographic research expeditions and cruises. We will leverage an extended series of workshops (both online and in-person), which have previously achieved significant progress and broad community acceptance in the 1st phase. The metadata scheme will build on the international standards from Phase 1, incorporating geo-location data and persistent identifiers such as DOIs for literature, ORCIDs for authors/researchers, IGSNs for samples, and RORs for research institutions, along with methodological catalogs. This collaborative effort in metadata standardization could result in defining heat flow as an essential ocean variable (EOV), thereby increasing the observational capacity of heat flow at the geosphere-hydrosphere interface.

Additionally, we will **develop protocols and (meta)data schemas for temperature as an observational parameter** measured with marine fiber optic data cables. This will be interconnected with the Helmholtz SAFAtor Large Research Infrastructure (*SMART Cables And Fiber-optic Sensing Amphibious Demonstrator; 2025–2029*). Building on Phase 1, we will continue and **complete** the **community-driven recompilation** and **metadata enrichment** of the global marine heat-flow data. This includes filling in missing information from the primary literature, defining flags for data quality and status, and adding additional selector variables for the GUI (WP1+2). We will also integrate data from approximately 150 publications not yet included in the GHFDB, primarily non-digitized international cruise reports from 1960 to 1990.

To address its limited availability, **historical cruise reports** from marine expeditions with heat-flow data will be collected, digitized, and released as digital copies, where copyright status and licenses allow **republication with DOI**. This will be done in cooperation with FID GEO and similar institutions, like Lamont-Doherty Earth Observatory and Woods Hole Oceanographic Institution. Released reports will be interlinked with the primary data. As in WP4, numerous volunteers from the geoscience and oceanography communities (see Section 5.3) will tackle this massive workload in a large international collaboration. Details of the organization of this collaborative approach are provided in WP5. Besides implementing and organizing the **collaboration between geoscience and oceanography**, our marine evaluation work will continue based on experiences with continental data over the past years. Well-documented, frequent releases of the database, similar to those in Phase 1 in 2023 and 2024, will be provided for download and shared in international trusted data repositories.

Tasks (responsible project partners)

T4.1 – Monitoring and evaluation of the Oceanography community expectations and requirements; development and maintenance of the collaborative community approach (UB + GFZ)

T4.2 – Develop concepts and metadata profiles, quality classes, and data classification for heat flow as an observational variable in marine subsea cables (UB, GFZ + collaborative network)

T4.3 – Data collection and metadata enrichment for marine heat flow measurements, including the extension of reference, author, and institution sub-database with persistent identifiers (UB + collaborative network)

T4.4 – Digitization/republication of historical marine heat flow publications/cruise reports (UB)

WP Deliverables

D4.1 – New metadata and quality classification for thermal properties and temperature data (M18-M36)

D4.2 – Release literature files with an open license (M24-M36)

WP5	Start–End: **M1–M36**
Project Management, Community Care and Outreach	
Lead: GFZ (Sven Fuchs), involved: all partners	

Description: This work package aims to coordinate and execute the scientific program, overseeing the planning, execution, and follow-up of the scientific activities outlined in WP1–WP4. It will also manage the recruitment process, as well as oversee the project's timetable and financial resources to ensure efficient and effective use of funds. Another objective is to facilitate and manage the collaborative efforts of the international community involved in the project, ensuring effective communication with various stakeholders to keep them informed

and engaged. The work package will also focus on organizing and implementing events and activities to maximize the project's visibility and impact, including scientific publications, public presentations, and other outreach events. Additionally, the work package aims to ensure that the project's results are effectively utilized and shared with the scientific community, the public, and stakeholders. The goal is to enhance the scientific impact on geoscientific research, increase public awareness, and boost the visibility of the project's findings. Proven tools, workflows, and the administrative support of GFZ staff will be utilized to achieve these objectives. For details on monitoring project progress, refer to Section 2.3.
Tasks (responsible project partners) T5.1 – Coordination of the consortium agreement (GFZ) T5.2 – Dissemination of scientific results (all) T5.3 – Communication and public engagement (GFZ) T5.4 – Monitoring and controlling of project progress (GFZ) T5.5 – Coordination of meetings & reports (all)
WP Deliverables D5.1 – Consortium agreement (M1) D5.2–5.4 – Workshops organized and held (M6–35) D5.5 – Scientific conferences attended (M12+) D5.6 – Final report (M36)

Global Deliverables

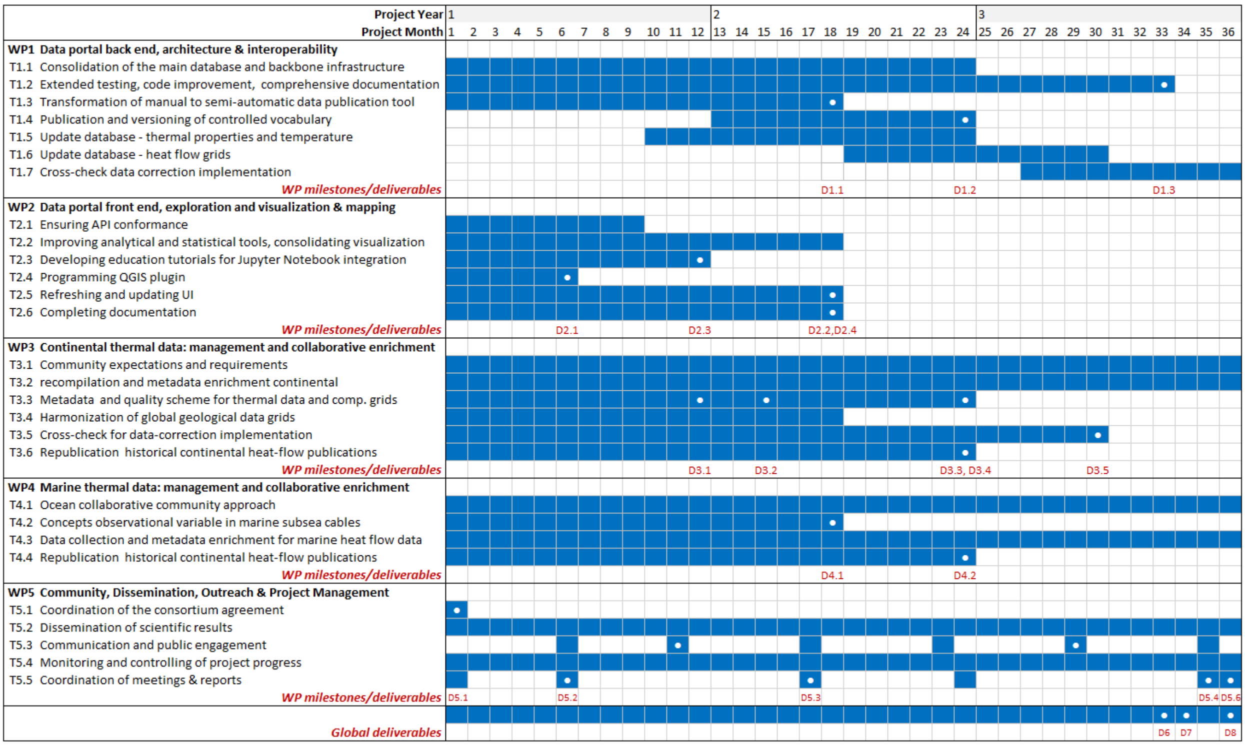
Besides the work package-specific deliverables, we have defined global deliverables where different partners and work packages contribute to:

- D6 – Full global heat flow dataset updated (M33)
- D7 – Final release of data portal operating (M34)
- D8 – Final project report (M36)

Risks and Risk Mitigation

No significant risks are expected for any of the planned work packages or technical tasks. Most of the planned developments are supported by extensive experiences from previous projects. Since the core team of Phase 1 intends to continue its participation, the risk in the **timely recruitment** of staff with appropriate programming skills is considered low. In addition, researchers on fixed-term contracts with the necessary skills from partner institutes could be invited to join the 2nd project phase. Additionally, training programs will be implemented to **upskill** existing staff. From phase 1, we have learned that the **collaborative reassessment** of the global heat-flow data is more time-consuming than anticipated. To address this in Phase 2, which corresponds to achieving the global deliverable D6, we are allocating a higher personal **budget** and seeking further third-party funding through external proposals (as done in Phase 1), and benefiting from the IHFC fellowship program. The experiences of the previous years have shown that many international heat-flow experts are supporting us. Frequent and open communications, maintaining the community spirit, and joint publications of intermediate results shall help to motivate researchers to complete the required work on time. To mitigate **technical risks**, we will implement robust security protocols and ensure system integration through pilot tests. Regular financial reviews and a contingency budget will address potential budget overruns. Stakeholder engagement and clear agreements with partners will mitigate collaboration risks. Compliance with data protection regulations and ethical guidelines will be ensured through the integration of the legal department in status project meetings. We will maintain a **risk register** to document and manage identified risks, and regular risk assessment meetings will be held to update strategies as necessary. This comprehensive approach aims to ensure the project's success and sustainability.

Gantt chart with project workflow



Monitoring of project progress

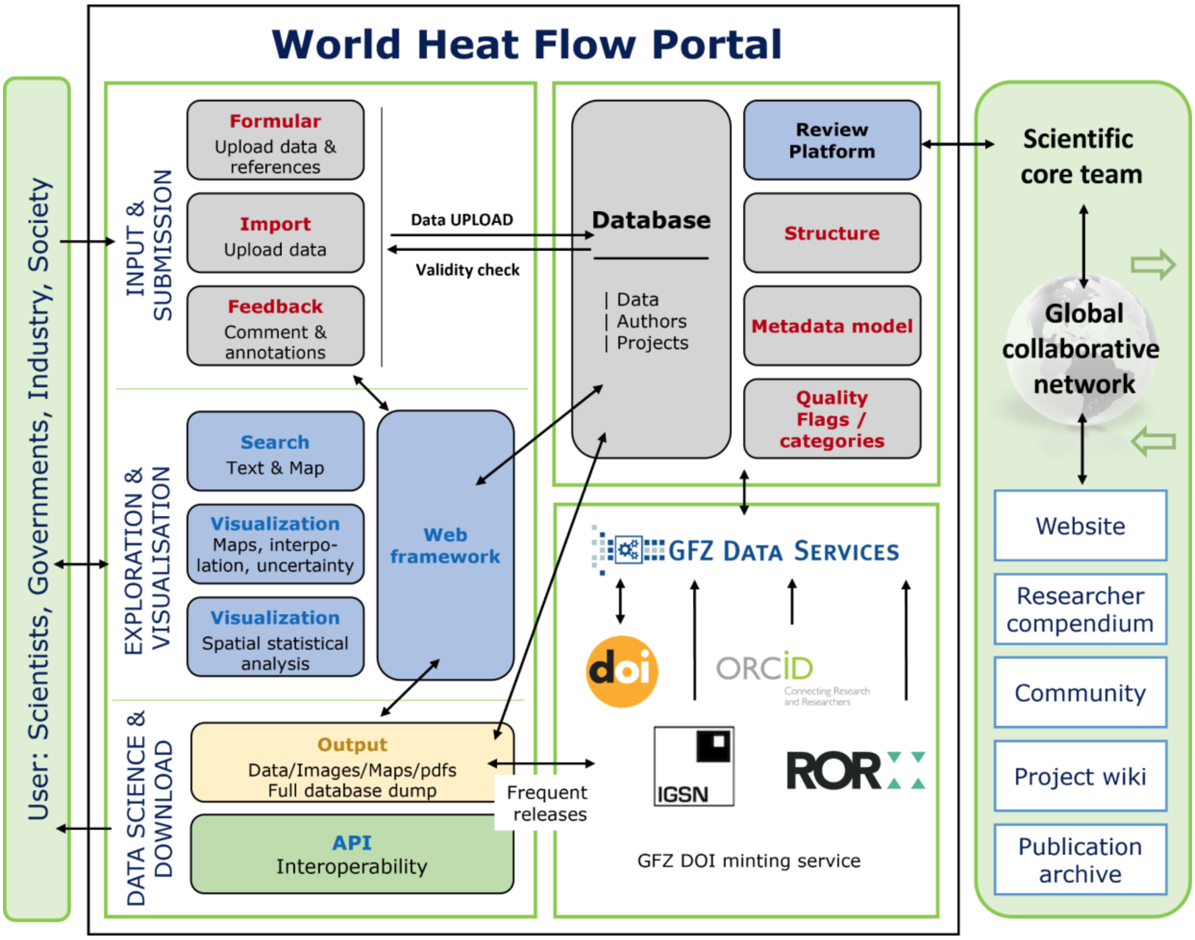
Biweekly video conferences (zoom) and bimonthly in-person two-day project workshops are important steering tools to ensure the project's progress. The iterative data revision of the international heat-flow data, performed by local and regional specialists (cf. Section 5), will be monitored by frequent quantitative statistics on the number of revised and updated datasets. For consistent planning of the data revision, quarterly statistics on the ongoing work will be calculated by the project lead in dialogue with all committed international supporters. This will provide the foundation for continuous adjustments and necessary focus shifts within the collaborative aspects of the project. We will utilize Helmholtz's Nextcloud software for project management) to track goals and monitor work progress according to the proposed schedule (see below, section 2.4.3). GitHub will be employed for version control and for tracking code changes, while online surveys and feedback forms will be used to collect feedback from stakeholders.

Outreach & General public-relations work

We will use various communication channels to convey the project's mission and progress to both the public and the science community. The main channel for dissemination to network members and external users will be the project website (reserved domain: www.heatflow.world). This website will be actively advertised on external websites and portals to ensure widespread dissemination of the project's results. The website will feature: (i) a comprehensive overview of the project, its members, and objectives (for experts and the general public); (ii) a public section where science publications, outreach material, announcements of workshops and meetings, shared data/software and other resources will be made available to the science community; (iii) a restricted area for consortium members to share documents, data, and tools. It will also host updates on recent developments, activities, and informal exchanges between the project and the scientific community and link to the heat flow data portal. The website will be hosted at the GFZ IT-Department and maintained by the project team.

Complementing the website, an annual email newsletter published by the GFZ group will provide an additional proactive way to communicate the project activities to members and the public. A brochure will be produced, in an accessible language, for distribution at local events, conferences, and public meetings. Beyond the website and newsletter, we will actively use social media channels to keep the scientific and general public informed about our activities and results. Key channels are YouTube (channel: Thermal Geophysics), ResearchGate, LinkedIn, GitHub, and similar online platforms. To maximize public awareness, we will engage with national and international stakeholders and entities (e.g., IASPEI, IUGG, EGA, EGEC, DFG) through press releases and educational outreach initiatives.

Structural overview



3. Project- and subject-related list of publications (PI selected 10)

Balkan-Pazvantoğlu et al. (2024): Update of the Heat Flow Database in Türkiye. doi.org/10.5880/GFZ.4.8.2024.001

Bernard, L.; et al. (2014): Scientific geodata infrastructures: challenges, approaches and directions. *International Journal of Digital Earth* 7(7): 613–633. https://doi.org/10.1080/17538947.2013.781244

Conze, R., et al. (2017): Utilizing the International Geo Sample Number Concept in Continental Scientific Drilling During ICDP Expedition COSC-1. - *Data Science Journal*, 16, 1, 1–8. https://doi.org/10.5334/dsj-2017-002

Elger, K., Bertelmann, R., Ulbricht, D., Haberland, C., Lauterjung, J., & Radosavljevic, B. (2018). FAIR am GFZ. System Erde, 8(1): 8, pp. 52-59. https://doi.org/10.2312/GFZ.syserde.08.01.8

Eppink, F.V.; et al. (2012): Land management and ecosystem services: How collaborative research programs can support better policies, *GAIA-Ecological Per. for Sc. and Society*, (21): 55–63. https://doi.org/10.14512/gaia.21.1.14

Förster, A., et al. (2016). Exploration of the enhanced geothermal system (EGS) potential of crystalline rocks for district heating (Elbe Zone, Saxony, Germany). *Int. J. of Earth Sci.*, 1–13. https://doi.org/10.1007/s00531-016-1429-6

Förster, A., Fuchs, S., Förster, H.-J., Norden, B. (2021): Ambiguity of crustal geotherms: A thermal-conductivity perspective. - *Geothermics*, 89, 101937. https://doi.org/10.1016/j.geothermics.2020.101937

Fuchs, S., et al. (2015). Calculation of thermal conductivity, thermal diffusivity and specific heat capacity of sedimentary rocks using petrophysical well logs. *GJI*, 203(3), 1977–2000. https://doi.org/10.1093/gji/ggv403

Fuchs, S., et al. (2020). Deep basin temperature and heat-flow field in Denmark – New insights from borehole analysis and 3D geothermal modelling. *Geothermics*, 83, 101722. https://doi.org/10.1016/j.geothermics.2019.101722

Fuchs, S., et al. (2021a) A new data-base structure for the IHFC Global Heat Flow Database. *IJTHFA*, 4(1), https://doi.org/10.31214/ijthfa.v4i1.62

Fuchs, S.; Norden, B.; International Heat Flow Commission (2021b): The Global Heat Flow Database: Release 2021. *GFZ Data Service*s. https://doi.org/10.5880/fidgeo.2021.014

Frick, M., Kranz, S., Norden, B., Bruhn, D., Fuchs, S. (2023): Dataset to Geothermal Resources and ATES Potential of Mesozoic Reservoirs in the North German Basin. https://doi.org/10.5880/GFZ.4.8.2023.004

Fuchs, S., et al. (2023): Quality-assurance of heat-flow data: The new structure and evaluation scheme of the IHFC Global Heat Flow Database. - *Tectonophysics*, 863, 229976. https://doi.org/10.1016/j.tecto.2023.229976

Fuchs, S., & Balling, N. (2016a). Improving the temperature predictions of subsurface thermal models by using high-quality input data. Part 1: Uncertainty analysis of the thermal-conductivity parameterization. *Geothermics*, 64, 42–54. http://doi.org/10.1016/j.geothermics.2016.04.010

Fuchs, S., & Balling, N. (2016b). Improving the temperature predictions of subsurface thermal models by using high-quality input data. Part 2: A case study from the Danish-German border region *Geothermics*, 64, 1–14. http://doi.org/10.1016/j.geothermics.2016.04.004

Fuentes-Bustillos, K., Neumann, F., Norden, B., Fuchs, S. (2023): Update of the Heat Flow Database in Mexico and surrounding areas. https://doi.org/10.5880/FIDGEO.2023.032

Global Heat Flow Data Assessment Group, et al. (2023): The Global Heat Flow Database: Update 2023. *GFZ Data Services* https://doi.org/10.5880/fidgeo.2023.008

Global Heat Flow Data Assessment Group, et al. (2024): The Global Heat Flow Database: Release 2024. *GFZ Data Services* https://doi.org/10.5880/fidgeo.2024.014

Henzen, C.; Mäs, S.; Bernard, L. (2013): Provenance Information in Geodata Infrastructures. Geographic Information Science at the Heart of Europe. *Springer*: 131–151. https://doi.org/10.1007/978-3-319-00615-4_8

Henzen, C.; et al. (2014): GeoMetaFacet 2.0 - Interaktive nutzerfreundliche Visualisierung von geographischen Metadaten. *Geoinformatik* 2014.

Henzen, C. et al. (2016): A Scientific Data Management Infrastructure for Environmental Monitoring a. Modelling. Proc. iEMSs Conf. 2016, Toulouse.

Henzen, C. (2018a): Building a Framework of Usability Patterns for Web Applications in Spatial Data Infrastructures. In: *International Journal of Geo-Information*, 11 (7), 2018. ISPRS, pp. 446. https://doi.org/10.3390/ijgi7110446

Henzen, C. (2018b): Usability von Webanwendungen in Geodateninfrastrukturen. *gis.Science*, (4), 2018. Wichmann V., pp. 133–143.

Jessop, A. M., et al. (1976). The World Heat Flow Data Collection - 1975. Geological Survey of Canada, Earth Physics Branch, *Geothermal Series*, 5, 10. https://doi.pangaea.de/10013/epic.40176.d002

Kindling, M., et al. (2017) The landscape of research data repositories in 2015: A re3data analysis. *D-Lib Magazine*, 23 (3–4).

Klump, J., et al. (2006) Data publication in the open access initiative. *Data Science Journal*, 5, pp. 79–83.

Norden, B., et al. (2020). Temperature and pressure corrections applied to rock thermal conductivity: impact on subsurface temperature prognosis and heat-flow determination in geothermal exploration. *Geothermal Energy*, 8(1), 1–19. https://doi.org/10.1186/s40517-020-0157-0

Norden, B., Förster, A., & Fuchs, S. (2023). The quality assessment of the German heat flow database. In XXVIII General Assembly of the (IUGG). https://doi.org/10.57757/IUGG23-3270.

Pampel, H., Weisweiler, N. L., Strecker, D., Witt, M., Vierkant, P., Elger, K., Bertelmann, R., Buys, M., Ferguson, L. M., Kindling, M., Kotarski, R., & Petras, V. (2023). re3data – Indexing the Global Research Data Repository Landscape Since 2012. In *Scientific Data* 10, 1, https://doi.org/10.1038/s41597-023-02462-y

Pfender, M. and Villinger, H., (2002) Miniaturized data loggers for deep sea sediment temperature gradient measurements, *Marine Geology*, 186 (3–4), 557-570, https://doi.org/10.1016/S0025-3227(02)00213-X.

Sidigam, E. R. et al. (2023): Update of the Heat Flow Database in India. https://doi.org/10.5880/FIDGEO.2023.038

Stall, S., et al. (2018): Advancing FAIR data in Earth, space, and environmental science. - *EESS News*, 99. /10.1029/2018EO109301

Stegmann, S. et al. (2006) Design of a modular, marine free-fall cone penetrometer. *Sea Technology*, 2, 27-32.

Ulbricht, D., et al. (2016) PanMetaDocs, eSciDoc, and DOIDB-An infrastructure for the curation and publication of file-based datasets for GFZ Data Services. *ISPRS International Journal of Geo-Information*, 5 (3), 25

Wiemann, S.; et al. (2018). Ad-hoc combination and analysis of heterogeneous and distributed spatial data for environmental monitoring - design and prototype of a web-based solution. *International Journal of Digital Earth* 11(1): 79–94. https://doi.org/10.1080/17538947.2017.1326984

Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. Sci Data 3, 160018 (2016). https://doi.org/10.1038/sdata.2016.18

4. Supplementary information on the project context

4.1 General ethical aspects

We do not anticipate any risks, harm to individuals or groups, or potential negative effects arising from our proposed research. Our project is neither categorized as Dual Use Research of Concern (DURC) nor does it have trade relevance. International cooperation is a cornerstone of our data assessment process. While open access to shared data could theoretically be misused, such as by publishing relevant data separately, we mitigate this risk through stringent data management and transparency measures. For over four years, we have maintained an open and transparent communication process with the entire community, ensuring that all actions are publicly documented. This approach has successfully prevented misuse, as the potential loss of reputation makes such actions unattractive. To safeguard the integrity and security of our technological developments, including the code of the infrastructure, access is restricted to project partners and approved scientific guests. This controlled access ensures that our developments are used responsibly and ethically. For the community-relevant parts, we also prioritize data protection and compliance with legal frameworks such as the General Data Protection Regulation (GDPR). Our infrastructure incorporates robust encryption and anonymization techniques to protect personal data and prevent unauthorized access. We promote responsible data sharing by providing open-access releases with clear documentation and data provenance, facilitating transparency and reproducibility in research. Regarding project staffing, we anticipate the continued involvement of key personnel, including the main developer and postdoctoral researchers from WHDB (Phase 1), ensuring consistency and expertise in our work.

4.2 Considerations on aspects of ecological sustainability in the planning and implementation of the project

Our research project is committed to ecological sustainability throughout its planning and implementation phases, including key aspects like **energy efficiency, sustainable software practices and carbon footprint.** We utilize the GFZ energy-efficient data center with reuse of energy for the campus heating systems to minimize energy consumption. We will also use virtualization to maximize resource use. Like in the 1st phase, we will adopt green software engineering principles to ensure efficiency across the software lifecycle. Where possible, we will reuse code from previous infrastructure projects of the partners. The GFZ data center employs data compression, deduplication, and tiered storage strategies to minimize the physical storage footprint. Proper disposal and recycling of obsolete equipment will be ensured to reduce environmental impact. Like in phase 1, we use remote video-conferencing tools to reduce the need for physical travel and related emissions. During our engagement with our community and collaborative network, we promote a broader adoption of these practices. Through these measures, we aim to create a sustainable and environmentally friendly research data infrastructure, ensuring that our project contributes positively to ecological sustainability.

4.3 Measures to meet funding requirements and handle project results

The project partners have been operating open-science data services (GFZ Data Service, PANGAEA) and global research infrastructures (WSM, GEOFON, GNSS, etc.) for years and have gained experience in developing and implementing these community services successfully. Similar to the proposed project, re3data.org stands as one example of a service that has been developed in Germany and is nowadays an internationally recognized scientific resource and web portal. Initially started as a third-party-funded project, it became part of the regular and permanent operations of the GFZ to ensure their sustainability. A similar long-term sustainability concept is developed for the proposed global heat-flow data infrastructure (cf. next section).

Concrete actions that will be taken after funding has expired are listed in 4.3.1. All project partners support the networking and interconnection of the project within the global geoscientific community. As members of global networks, like IHFC and ILP (cf. 1.3), we benefit from a well-established collaborative network. As in the 1st phase, all the findings of the project will not be only announced to the relevant geoscientific community (cf. WP5), but will be made also available to third parties free of charge, e.g., with an API function. As we use open-access software packages for the programming work, the source code produced will be made available open source. Again, we will actively publish our results in peer-reviewed journals (open access priority) and present them at international conferences to ensure broad dissemination and impact within the scientific community.

Permanent commitment and sustainability plan

With the presented concept, we propose initial financing for a long-term nationwide RDI of international importance. As one of the largest geoscientific research centers in Europe, **GFZ is able and committed to maintaining and safeguarding the project findings and ensuring their sustainability** (cf. attached GFZ commitment letter).

The system will be driven and maintained by the Section Geoenergy (data curation) with the support of the IT department (technical maintenance of server) and the GFZ LIS section (administrative and technical maintenance of database programming) of the GFZ. A **permanent position** in the Geoenergy section will sustain the long-term professional maintenance of the database (data curation and management) after funding expires (cf. section 5.1). The broad and long-term experience of the members of the LIS Section (where the project PI Kirsten Elger has a permanent contract) will **ensure** that **any relevant technical standards** particularly those concerning data collection and data storage are considered during the establishment, enhancement and operation of the RDI. This infrastructure is **designed for the needs** of the community, measured by a thorough analysis of several surveys conducted in the run-up of this project. We will ensure that the infrastructure is scalable and adaptable to future growth and new technologies. These **requirements will be monitored** in the future by collecting feedback and continuing our activities in national (German Geophysical Society DGG) and international boards (e.g., International Heat Flow Commission IHFC). Based on his monitoring we will adapt our activities if required. We will also provide ongoing training and support to users to ensure effective use and integration of the infrastructure into their research workflows. Core elements of the new service, like the interoperability with DOI, ORCID, and IGSN systems, are ensured as these data services are **already implemented** in the permanent structure of **GFZ Data Services** and the IT department. The new infrastructure will be interlinked and integrated with the offerings of the German NFDI and, in particular, NFDI4Earth. We will establish clear governance structures that define roles, responsibilities, and decision-making processes. Through these comprehensive measures, we aim to create a sustainable and impactful RDI that serves the scientific community effectively and responsibly. Our commitment and sustainability plan will benefit from the many experiences **GFZ has collected with similar permanent global** RDI and services, like the World Stress Map (WSM) the global seismic network (GEOFON), geomagnetic observatories, GNSS Station Network or the platforms for the GFZ satellite missions CHAMP, GRACE and GRACE-FO.

Scientific and financial involvement of international cooperation partners

The collaborative part of the proposed project will be conducted in **close cooperation with an international network** of heat-flow experienced colleagues that was built and grown during Phase 1 of this project. Among them are members of the International Heat Flow Commission

(IHFC) of the International Association of Seismology and Physics of the Earth's Interior (IASPEI/IUGG) and of the International Lithosphere Program (ILP) of IUGG and IUGS. The IHFC members, currently 20 researchers from 5 continents with long experience in heat-flow research, will constitute the backbone of geoscientists that bear the international collaborative revision of the database. **None of them applied for individual funding** related to this proposal yet. As far as fellows of the IHFC support the assessment work like in Phase 1, they support the progress in WP 3 and 4 backed with travel support from the IHFC. Although the proposed project is **not set up as an official** part of ongoing **agreements** between the DFG and other partner organizations, some of the collaborating researchers are located at such institutes. The IHFC members will also act as multiplicators towards the broader geoscientific community, related exploration industry, and other stakeholders to maximize the public awareness of this project. Amongst others, **support letters** of NFDI4Earth IHFC, IASPEI, IUGG, KIT and LMU are attached to this proposal.

4.4 Formal assurances

We declare:

"Publications resulting from the project and any relevant documentation will be available via open access, making them widely accessible for use by third parties. Furthermore, the source code for the software developed under the project will be documented in accordance with established standards, licensed with an open-source license, and made available for use by third parties free of charge."

5. People/collaborations/funding

5.1 Employment status information

GFZ-LIS

Elger, Kirsten Dr. — Scientist, permanent contract, institutional funding

GFZ-ETG

Fuchs, Sven Dr. — Postdoc and working group leader, fixed term contract with institutional funding (current contract until 12/2024); start of change of employment status to permanent position in 2025/26

TUD: Mäs, Stephan Dr. — Scientist, permanent contract, institutional funding

UB: Kopf, Achim Prof. — Scientist, permanent contract, institutional funding

5.2 Composition of the project group

GFZ-LIS

Elger, Kirsten Dr. — Scientist (see above), 10%

Frenzel, Simone — Programmer, Section Library and Information System (Telegrafenberg Library), permanent contract, 20%

GFZ-ETG

Fuchs, Sven Dr. — Scientist (see above), 20%

Norden, Ben Dr. — Scientist, Section Geoenergy – permanent contract, 20%

Elbarbary, Samah Dr. — Scientist, Section Geoenergy – fixed contract, 10%

TUD: Mäs, Stephan Dr. — Scientist, Chair of Geoinformatics, permanent contract, institutional funding, 10%

UB: Kopf, Achim Prof. — Scientist, Research Group Marine Geotechnics, permanent contract, institutional funding, 5%

5.3 Institutions or researchers in Germany (agreed to cooperate on this project)

Agreements for national collaborations within our project has been made with:

Technical University of Darmstadt, Karlsruhe Institute for Technology, NFDI4Earth, University of Bremen/MARUM, Fraunhofer-Einrichtung für Energieinfrastrukturen und Geothermie IEG

5.4 Institutions or researchers abroad (agreed to cooperate on this project)

A selection of scientists that committed on a voluntary level to support the collaborative revision of the database. Many of them are connected since the beginning of Phase 1:

Global:	International Heat Flow Commission (**IHFC**) of the IASPEI/IUGG Task Forces of the International Lithosphere Program (**ILP**) of IUGG/IUGS
Africa:	Mohammed Hichem Bencharef (Echahid Cheikh Larbi Tebessi University, **Algeria**), Insaf Mraidi (University of Tunis El Manar, **Tunisia**)
Asia:	Labani Ray (**India**), Shaowen Liu (Nanjing University, **China**), Guangzheng Jiang (**China**), Akiko Tanaka (Geological Survey of **Japan**)
Australia:	Graeme Beardsmore (Melbourne), Tobias Stål (University of Tasmania), Lesley Wyborn (The Australian National University)
Europe:	Massimo Verdoya (University of Genova, **Italy**), Jeffrey Poort (Sorbonne University & CNR, **France**), Niels Balling (Aarhus University, **Denmark**)
North America:	Robert Harris (Oregon State University, **USA**), Raquel Negrete-Aranda (CICESE – CONAHCYT, **Mexico**)
South America:	Suze Pereira Guimarães (Geosciences Department at Federal Rural University of Rio de Janeiro, **Brazil**), Jhon Camilo Matiz Leon; Harold Buitrago (Colombian Geological Survey, **Colombia**)

5.5 Institutions and researchers (collaborated scientifically within the past 3 years)

GFZ-LIS:

National: KIT (Thomas Forbriger, Hans-Jürgen Göbelbecker); Institut für Bibliotheks- und Informationswissenschaft (Vivien Petras); ; Niedersächsische Staats- und Universitätsbibliothek Göttingen (Malte Semmler, Inke Achterberg); UFZ (Thomas Schnicke, Jan Bumberger), TU Dresden (Lars Bernard). *International*: British Geological Survey (Keyworth: Helen Glaves, Helen Taylor, Edinburgh: Simon Flower/ INTERMAGNET), CSIRO (Jens Klump), Utrecht University (The Netherlands: Richard Wessels, Geertje ter Maat, Otto Lange), INGV (Rome, Italy: Massimo Cocco, Aldo Winkler), Universita Roma Tre (Rome, Italy: Francesca Funiciello), Politecnico Milan (Mirko Reguzzoni), American Geophysical Union (Shelley Stall), CSIC (José-Luis Fernandez-Turiel), ETH Zürich (Alba Zappone, Florian Haslinger), DataCite (u.a. Matt Buys, Rorie Edmunds, Cody Ross, Kelly Stathis)), University of Columbia/ Lamont Doherty Earth Observatory (Kerstin Lehnert/ IEDA), Geosciences Australia/ NCI (Lesley Wyborn)

GFZ-ETG:

National: KIT (Thomas Kohl); University of Bremen (Norbert Kaul), Ruhr University Bochum (Christophe Pascale, Rodolfo Christiansen), Geomar Helmholtz Centre for Ocean Research Kiel (Michael Riedel), Friedrich-Schiller-Universität (Roman Esefelder)
International: Aarhus University (Denmark; Niels Balling); GEUS, Copenhagen (Denmark, Anders Mathiesen, Lars-Henrik Nielsen, William Colgan); Geological Survey of Japan (Japan, Akiko Tanaka), University of Helsinki (Finland, Ilmo Kukkonen), National Research Council (Italy, Gianluca Gola), Czech Academy of Sciences (Czech Republic, Jan Safanda), Universidad Michoacana de San Nicolás de Hidalgo (México, Orlando Miguel Espinoza-Ojeda), Geological and Mining Institute of Spain IGME-CSIC (Spain, Ignazio Marzan), ETH Zürich (Swiss, Ladislaus Rybach), Dokuz Eylül University (Türkiye, Elif Balkan-Pazvantoğlu), Laboratório Nacional de

Energia e Geologia (Portugal, Elsa Cristina Ramalho), Sorbonne Université, CNRS (France, Jeffrey Poort, Frédérique Rolandone), Chinese Academy of Sciences (China, Yibo Wang, Xiang Gao, Shengbiao Hu), University of Tartu (Estonia, Argo Jõeleht), Geological Survey of Slovenia (Slovenia, Dušan Rajver), Nanjing University (China, Shaowen Liu), Oregon State University (USA, Robert Harris), SMU Geothermal Laboratory (USA, Maria Richards), University of Melbourne (Australia, Graeme Beardsmore, Sandra McLaren, Belay G. Mino, Florian Forster), University of Genoa (Italy, Massimo Verdoya, Paolo Chiozzi), Chevron Technical Center, Houston (USA, Jeffrey Nunn), GNS Science (New Zealand, Rob Funnell), Prince of Songkla University (Thailand, Helmut Duerrast), University of Perugia (Italy, Christina Pauselli), Centro de Investigacion Cientifica y de Educacion Superior de Ensenada (Mexico, Raquel Negrete-Aranda; Karina Fuentes), Dublin Institute for Advanced Studies (Ireland, Emma Chambers), CSIR National Geophysical Research Institute (India, Eswara Rao Sidagam), Universidad Nacional de Colombia (Colombia, Jhon Camilo Matiz-Leon), Echahid Cheikh Larbi Tebessi University (Algeria, Mohammed Hichem Bencharef), The University of Texas at Austin (USA, Mohamed Shafik Khaled), Universidade de Evora (Portugal, Maria Rosa Alves Duque), The Ohio State University (USA, Martina Leveni), University of Tasmania (Australia, Tobias Staal), Universidad Autónoma de Nuevo León (Mexico, Ana Paulina Anguiano Dominguez), AGH University of Science and Technology (Poland, Marek Hajto), Malaysian Continental Shelf Project, National Security Council (Malaysia, Mazlan Madon), Geological Survey of Israel (Israel, Itay J. Reznik), University of California (USA, Jyun-Nai Wu), Université de Lorraine (France, Audrey Bonnelye), Institut de Radioprotection et de Sûreté Nucléaire IRSN (France, Pierre Dick), China University of Geosciences (China, Ruyang Yu, Shu Jiang, Peng Peng, Yuanping Li, Hu Wang), University Centre in Svalbard (Norway, Kim Senger, Matthijs Nuus, Peter Betlem, Tom Birchall, Hanne H. Christiansen), Geological Survey of Norway (Norway, Harald Elvebakk), Store Norske Energi AS (Norway, Malte Jochmann),

TUD:
National: UFZ Leipzig (Ralf Seppelt), HTW Dresden (Frank Schwarzbach), LMU München (Wolfram Mauser, Florian Zabel), iDiv Uni Leipzig (Carsten Mayer), 52°North (Albert Remke, Simon Jirka), HS Beuth Berlin (Petra Sauer), Kiel Earth Institute Christian-Albrechts-Universität zu Kiel (Gernot Klepper, Ruth Delzeit); Potsdam Institut für Klimafolgenforschung (Alexander Popp)
International: Universität Salzburg (Barbara Hofer), IIT Mandi India (Dericks Shukla), University of Free State South Africa (Marinda Avenant, Maitland Seaman), North West University South Africa (Adeline Ngie), University of Pretoria South Africa (Serena Coetzee)

UB:
National: GEOMAR Kiel (Eric Achterberg, Christian Berndt, Heidrun Kopp, Andreas Oschlies, Lars Rüpke, Klaus Wallmann), Univ. Hamburg (Christian Hübscher, Beate Ratter), IOW Warnemünde (Gregor Rehder, Oliver Zielinski), Univ. Hannover (Torsten Schlurmann), Fraunhofer IPM Freiburg (Raimund Brunner), RWTH Aachen (Klaus Reicherter), AWI Bremerhaven (Katja Metfies, Boris Dorschel), ZMT Bremen (Martin Zimmer).
International: Univ. Innsbruck (Michi Strasser), JAMSTEC (Eiichiro Araki, Masa Kinoshita), Oregon State University (Rob Harris), Univ. of Texas in Austin (Demian Saffer), Ocean Networks Canada (Kate Moran, Martin Scherwath), CARBFIX Iceland (Martin Voigt), NOCS (Jurg Matter), Virginia State University (Nian Stark), GNS New Zealand (Agnes Reyes, Rupert Sutherland), IFREMER France (Nabil Sultan, Jan Opderbecke), NIOZ (Henk Brinkhuis).

5.6 Project-relevant cooperation with commercial enterprises

- not applicable -

5.7 Project-relevant participation in commercial enterprises

- not applicable -

5.8 Other submissions

- not applicable -

5.9 Financial contributions

The participating institutions provide some personal from the permanent staff support the work, controlling, and the management of the project (cf. section 5.1 and 5.2). On top, all project partners offer and provide their own infrastructural or financial contributions, like office space, equipment, costs of IT infrastructure and maintenance, and support with controlling and administration. In particular the costs of the operation of the project infrastructure is granted after the projects end. The purchase of large additional equipment is not required and direct funding from GFZ, TUD and/or UB to the project is not provided.

5.10 Requested modules/funds

Basic Module

For each applicant, we apply for funding within the Basic Module.

Applicant: Elger, Kirsten (GFZ-LIS)

Category	Applied funding	Description
Personal	**Postdoc** (Samuel Jennings), 36 PM, 100%, E13	Main developer for lead and work in WP1, support of WP2
Personal	**Postdoc** IT specialist (n.n.), 36 PM, 50%, E13	Development work for consolidation and work in WP1 and on IT infrastructure. Requires person with both research data management and programming skills.
Personal	**Student worker,** 36 PM, 40h/month (sum 31,285 €)	Help with tools & website programming, Start summer 2025
Travel costs	5,300 €	Covers travel costs for project meetings (1,000 €/year) and visits at conferences and scientific meetings, like DGGV/DGG Assembly, NFDI4Earth, RDA Plenary
Publication costs	2,200 €	For payment of publication fees for open-access journals. (750 € each)

Applicant: Fuchs, Sven (GFZ-ETG)

Category	Applied funding	Description
Personal	**Postdoc** (Florian Neumann), 36-person months (PM), 100%, E13	Person for overall project lead, lead and work in WP3 and WP5
Personal	**3 Student workers,** 36 PM each, 60h/month	Help with data screening and literature review and assessment, geoscientific background required,

	(sum 140,783.90 €)	Start M1
Travel costs	12,700 €	Covers travel costs for bi-monthly project meetings (5 meetings/year; 1,000 €/year), kick off and final project meeting, and visits at conferences and scientific meetings Annual DGG Assembly (500 €), WGC 2026 Canada (3,300 €), and IUGG 2027 Korea (5,900 €)
Guests	2,480 €	Covers visits of min. 2 incoming international guests at the GFZ. Assuming an average duration of 5 days, we estimate the cost of one visit at 700 € for traveling and 90 €/night for accommodation, summing up to 1,240 € per guest
Publication costs	2,250 €	For payment of publication fees for open-access journals. (750 € / year)

Applicant: Mäs, Stephan (TUD)

Category	Applied funding	Description
Personal	Scientist (Nikolas Ott), 18 PM, 100%, E13	Person for lead and work of WP2, support of WP1
Personal	**Student worker,** 12 PM, 40h/month (sum 10,121 €)	Help with tool programming and usability evaluation
Travel costs	5,250 €	Covers travel costs for project meetings (1,000 €/year) and visits at conferences and scientific meetings (2,250 €), like NFDI4Earth, RDA Plenaries, IScience or AGILE in 2025, 2026, 2027, IUGG 2027
Publication costs	2,250 €	For payment of publication fee for one open-access journal (750 €)

Applicant: Kopf, Achim (UB)

Category	Applied funding	Description
Personal	Scientist (n.n.), 36 PM, 66%, E13	Person for lead and work of WP3, support of WP2
Personal	**Student worker,** 18 PM, 40h/month (sum 15,295 €)	Help with literature review and assessment work, Start early 2025
Travel costs	5,250 €	Covers travel costs for bi-monthly project meetings (5 meetings/year; 1,000 €/year), kick off and final project meeting, and visits at conferences and scientific meetings (2,250 €), like Annual DGG Assembly, EGU 2026, IASPEI 2025
Publication costs	2,250 €	For payment of publication fees for two open-access journals. (750 € each)

Module: Project-specific workshops

The PI's apply for funding within the module "project-specific workshops".

Institution	Applied funding	Description
GFZ-ETG (Sven Fuchs)	11,250 €	Education workshop and summer school (2025): 1-week "Quality procedures and heat flow data treatment" Costs for travel & accommodation (6,000 €), organization (2,250 €), catering (2,000 €) and documentation (1,000 €), ca. 20–25 international participants.
UB (Achim Kopf)	7,750 €	Workshop (2026): 3-day workshop "Legacy marine heat flow data. Standards and outlook" Costs for travel & accommodation (6,500 €), organization (250 €), catering (500 €) and documentation (500 €), ca. 15 international participants.
GFZ-ETG (Sven Fuchs)	17,200 €	Event (2027/2028) 3-day conference and release event "The World Heat Flow Portal – Final Release" Costs for travel & accommodation (9,000 €), organization (3,500 €), catering (3,500 €) and documentation 1,200 €), ca. 65 international participants.
Sum without overhead: 36,200 €		